

## Unit I – Introduction to Data Science

- **What is Data Science?** An interdisciplinary field that uses algorithms, scientific methods, and systems to extract knowledge and insights from structured and unstructured data.
- **Why is Data Science important in modern industries?** It enables data-driven decision-making, optimizes business operations, personalizes customer experiences, and predicts future market trends.
- **What are the major applications of Data Science?** Healthcare (disease prediction), finance (fraud detection), e-commerce (recommendations), and logistics (route optimization).
- **Differentiate between Data Science and Information Science.** Data Science focuses on analyzing data and building predictive models. Information Science focuses on the storage, retrieval, and management of information systems.
- **What is Business Intelligence?** Technologies and strategies used by enterprises for the data analysis of business information to provide historical, current, and predictive views of operations.
- **Difference between Business Intelligence and Data Science.** BI looks backward to analyze what happened (descriptive), while Data Science looks forward to predict what will happen (predictive).
- **What are the different types of data?** Structured, semi-structured, and unstructured data.
- **Explain structured, semi-structured, and unstructured data.** \* *Structured:* Highly organized, easily searchable (e.g., SQL databases).
  - *Semi-structured:* Contains tags or markers to separate elements (e.g., JSON, XML).
  - *Unstructured:* No pre-defined format, difficult to search (e.g., text, videos, images).
- **What is data collection?** The systematic process of gathering observations or measurements to answer research questions or test hypotheses.
- **What is data wrangling?** The process of transforming and mapping raw data into a more readable, usable format for analytics.
- **Why is data preprocessing necessary?** Raw data is often incomplete, noisy, or inconsistent. Preprocessing cleans it, improving the accuracy of machine learning models.
- **What is data cleaning?** The process of fixing or removing incorrect, corrupted, incorrectly formatted, duplicate, or incomplete data within a dataset.
- **Explain data integration with example.** Combining data from different sources into a unified view. *Example:* Merging customer purchase history from a SQL database with their website clickstream data from server logs.
- **What is data reduction?** Techniques used to reduce the volume of data while maintaining the integrity of the original data (e.g., PCA, sampling).

- **Explain data transformation techniques.** Methods to convert data into appropriate formats, such as normalization (scaling data to a specific range) or aggregation (summarizing data).
- **What is data discretization?** The process of converting continuous data (like age) into discrete intervals or bins (like "18-25", "26-35").
- **What are missing values in datasets?** Data points that are absent due to collection errors or non-response. They are typically handled by deletion or imputation.
- **How can duplicate data affect analysis?** It can skew results, artificially inflate statistical significance, and cause machine learning models to overfit.
- **What are outliers?** Data points that differ significantly from other observations in the same dataset, often indicating experimental errors or rare events.
- **Explain the student academic performance preprocessing case study.** Handling missing grades, removing duplicate student records, and normalizing attendance scores to build a clean dataset capable of predicting final exam results.

## Unit II – Statistical Inference

- **Why is statistics important in Data Science?** It provides the mathematical foundation to summarize data, identify patterns, quantify uncertainty, and validate hypotheses.
- **Define measure of central tendency.** A single value that attempts to describe a set of data by identifying its central position.
- **Explain mean with formula.** The mathematical average of a set of numbers. Calculated as  $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$
- **What is median?** The exact middle value of a dataset when it is ordered from lowest to highest.
- **What is mode?** The value that appears most frequently in a dataset.
- **Define mid-range.** The arithmetic mean of the maximum and minimum values in a dataset.
- **Difference between mean and median.** The mean incorporates all values and is sensitive to outliers. The median only looks at the middle position and is robust against outliers.
- **What are measures of dispersion?** Metrics that describe how spread out or scattered the data points are.
- **Define range.** The difference between the highest and lowest values in a dataset.
- **Explain variance.** The average of the squared differences from the Mean, measuring how far a set of numbers is spread out.
- **What is standard deviation?** The square root of the variance. It measures the average distance of data points from the mean.
- **What is mean deviation?** The average of the absolute differences between each data point and the mean.
- **Why are dispersion measures important?** They indicate the reliability of the central tendency. High dispersion means high variability and less predictability.
- **State Bayes Theorem.** A formula calculating conditional probability:  $P(A|B) = \frac{P(B|A)P(A)}{P(B)}$
- **What are the applications of Bayes Theorem?** Spam email filtering, medical disease diagnosis, and sentiment analysis.
- **What is hypothesis testing?** A statistical method used to make decisions or inferences about a population parameter based on sample data.
- **Differentiate null hypothesis and alternative hypothesis.** The Null Hypothesis ( $H_0$ ) assumes no effect or relationship. The Alternative Hypothesis ( $H_1$ ) assumes there is an effect or relationship.
- **What is Pearson correlation?** A statistic that measures the linear correlation between two continuous variables.
- **What is the range of correlation coefficient?** Between -1.0 and +1.0.

- **Explain positive and negative correlation.** *Positive:* As one variable increases, the other increases. *Negative:* As one variable increases, the other decreases.
- **What is Chi-square test?** A statistical test used to determine if there is a significant association between categorical variables.
- **What is t-test?** A test used to compare the means of two groups to determine if they are significantly different from each other.
- **Difference between Chi-square test and t-test.** Chi-square evaluates categorical data (counts/frequencies), while a t-test evaluates continuous data (means).
- **What is sample hypothesis testing?** Testing a smaller subset of data (a sample) to draw conclusions about the entire population.
- **Explain statistical analysis using employee dataset.** Using t-tests to check if average salaries differ by gender, or correlation to see if years of experience map linearly to salary increases.

## Unit III – Data Analytics Life Cycle

- **What is Data Analytics?** The science of analyzing raw data to make conclusions about that information.
- **Explain the Data Analytics Lifecycle.** A structured methodology defining the phases of a data science project from conception to deployment.
- **List all phases of Data Analytics Lifecycle.** 1. Discovery, 2. Data Preparation, 3. Model Planning, 4. Model Building, 5. Communicate Results, 6. Operationalize.
- **What happens in the Discovery phase?** The team defines the business problem, establishes goals, formulates initial hypotheses, and identifies available data sources.
- **Explain the Data Preparation phase.** Involves extracting, cleaning, transforming, and loading data into an analytics sandbox. It is the most time-consuming phase.
- **What is Model Planning?** Determining the methods, techniques, and workflow to build the model. Involves feature selection and exploratory data analysis.
- **What is the purpose of Model Building?** Developing datasets for training, testing, and production, and executing the chosen machine learning algorithms.
- **Explain the Communication Results phase.** Evaluating if the model succeeded or failed based on business criteria, and presenting these findings to stakeholders.
- **What does Operationalize mean?** Deploying the final model into a production environment so it can be used for real-world decision-making.
- **Why is data preparation important?** "Garbage in, garbage out." High-quality data is essential; models cannot learn accurately from messy or incorrect data.
- **What are the challenges in model building?** Avoiding overfitting, selecting the correct algorithms, computational resource limits, and tuning hyperparameters.
- **What is the role of business understanding in analytics?** It ensures the technical solution actually solves the real-world problem and provides a return on investment.
- **Explain iterative nature of Data Analytics Lifecycle.** The cycle is rarely linear. Insights in later phases often require returning to earlier phases to get new data or tweak models.
- **What tools are commonly used in analytics lifecycle?** Python, R, SQL, Hadoop, Tableau, and Git.
- **What is exploratory data analysis?** Analyzing datasets visually and statistically to summarize their main characteristics before formal modeling.
- **What is feature selection?** The process of selecting the most relevant variables or columns from your dataset to build your model.
- **Explain deployment of analytics models.** Integrating a completed machine learning model into an existing software application to automate predictions.
- **What is the GINA case study?** A classic textbook case study demonstrating the analytics lifecycle, often involving optimizing an IT environment or classifying text/networks.

- **Why is communication important in analytics?** Because highly technical insights are useless unless non-technical business leaders can understand and act upon them.
- **Difference between model planning and model building.** Planning is the strategy (choosing algorithms, exploring data). Building is the execution (writing code, training the algorithm).

## Unit IV – Predictive Data Analytics with Python

- **Why is Python used in Data Science?** It has simple syntax, a massive open-source community, and powerful dedicated libraries for math, data manipulation, and machine learning.
- **Name important Python libraries used in Data Analytics.** Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn.
- **What is NumPy?** A library used for working with large, multi-dimensional arrays and matrices, along with high-level mathematical functions.
- **What is Pandas?** A library providing high-performance, easy-to-use data structures (like DataFrames) for data manipulation and analysis.
- **What is Matplotlib?** A comprehensive library for creating static, animated, and interactive data visualizations in Python.
- **What is Scikit-learn?** The premier open-source machine learning library for Python, featuring classification, regression, and clustering algorithms.
- **How do you remove duplicate data in Python?** Using the Pandas function: `dataframe.drop_duplicates()`.
- **What is data transformation?** Changing the format, structure, or values of data (e.g., standardizing numerical features using `StandardScaler`).
- **Explain mapping functions in Python preprocessing.** Using `.map()` or `.apply()` in Pandas to substitute or alter values in a column based on a predefined dictionary or function.
- **How are missing values handled?** By removing the rows/columns (`dropna()`) or by replacing them with the mean, median, or mode (`fillna()`).
- **Difference between descriptive, predictive, and prescriptive analytics.** Descriptive: What happened? Predictive: What will happen? Prescriptive: What should we do about it?
- **What is predictive analytics?** Using historical data, statistical algorithms, and machine learning to forecast future outcomes.
- **Explain descriptive analytics with example.** Summarizing raw data to understand past behavior. *Example:* A company's quarterly sales report.
- **What is prescriptive analytics?** Analytics that suggest courses of action based on predictions (e.g., Google Maps suggesting a faster route due to predicted traffic).
- **What are association rules?** Unsupervised learning techniques used to discover interesting relationships or co-occurrences between variables (e.g., Market Basket Analysis).
- **Explain Apriori Algorithm.** An algorithm used to find frequent itemsets in a database and generate association rules based on support and confidence thresholds.
- **What is FP-growth algorithm?** "Frequent Pattern growth" is an algorithm that finds frequent itemsets using a tree structure, bypassing the costly candidate generation step of Apriori.

- **Difference between Apriori and FP-growth.** Apriori scans the database multiple times and generates candidates (slower). FP-growth scans it twice and builds a tree (faster and more memory efficient).
- **What is regression?** A predictive modeling technique that investigates the relationship between a dependent (target) continuous variable and independent variables.
- **Explain Linear Regression.** Models the linear relationship between variables by fitting a straight line to the data points.
- **What is Logistic Regression?** A statistical model used for binary classification, predicting the probability that an instance belongs to a specific class.
- **Difference between Linear and Logistic Regression.** Linear regression outputs continuous numerical values. Logistic regression outputs probabilities mapping to categorical outcomes.
- **What is classification?** A supervised machine learning technique where the model predicts a categorical label for new input data based on past observations.
- **Explain Naïve Bayes classifier.** A probabilistic classification algorithm based on Bayes' Theorem, with a "naïve" assumption that all features are independent of each other.
- **What is Decision Tree algorithm?** A flowchart-like model where internal nodes represent feature conditions, branches represent outcomes, and leaf nodes represent final class labels.
- **What is overfitting?** When a model learns the training data *too* perfectly, including its noise, causing it to perform poorly on new, unseen data.
- **What is training data and testing data?** *Training data:* The subset used to teach the model. *Testing data:* The unseen subset used to evaluate the model's performance.
- **Explain preprocessing on IRIS dataset.** Checking for missing values, separating features (sepal/petal dimensions) from the target label (species), and encoding the species strings into numbers.
- **What are labels and features?** *Features:* The independent variables or inputs. *Labels:* The dependent variable or output you are trying to predict.
- **Explain regression using Scikit-learn.** Import LinearRegression from sklearn.linear\_model, instantiate the model, call .fit(X\_train, y\_train), and then generate predictions using .predict(X\_test).



## Unit V – Data Analytics and Model Evaluation

- **What is clustering?** An unsupervised learning technique that groups similar data points together into clusters without prior knowledge of labels.
- **Explain K-Means clustering.** An algorithm that partitions data into  $K$  distinct clusters by minimizing the distance between data points and the centroid of their cluster.
- **What is Hierarchical Clustering?** An algorithm that builds a hierarchy or tree of clusters (a dendrogram), either by merging smaller clusters or dividing larger ones.
- **Difference between K-Means and Hierarchical clustering.** K-Means requires you to predefine the number of clusters ( $K$ ) and is faster. Hierarchical does not require predefining  $K$  but is computationally expensive.
- **What is centroid in K-Means?** The imaginary or real location representing the exact center (mean) of the cluster.
- **Explain Elbow Method.** A heuristic used to determine the optimal number of clusters ( $K$ ) by plotting the Within-Cluster-Sum-of-Squares and looking for a "bend" or elbow in the graph.
- **What is time-series analysis?** The statistical analysis of data points recorded at specific, constant time intervals to identify trends or seasonal patterns.
- **Give applications of time-series analysis.** Stock market forecasting, weather prediction, and server load monitoring.
- **What is text analysis?** The process of automatically extracting machine-readable information, meaning, and sentiment from unstructured text.
- **Explain text preprocessing.** Cleaning text data by lowercasing, removing punctuation, removing stop words, and applying stemming or lemmatization.
- **What is tokenization?** The process of breaking down a text sequence into smaller units, such as words, subwords, or sentences (tokens).
- **What are stop words?** Extremely common words (e.g., "the", "and", "is") that are filtered out before processing because they carry little to no meaningful information.
- **Explain Bag of Words model.** A representation of text that counts the frequency of words in a document, ignoring grammar and word order.
- **What is TF-IDF?** Term Frequency-Inverse Document Frequency. It evaluates how important a word is to a specific document within a larger collection (corpus).
- **What is topic modeling?** An unsupervised statistical technique used to discover abstract "topics" or themes hidden across a collection of documents.
- **What is social network analysis?** The process of investigating social structures using networks and graph theory (nodes and edges).
- **What is business analytics?** The practice of iterative, methodical exploration of an organization's data with an emphasis on statistical analysis to drive business decisions.
- **What is model evaluation?** The process of using different evaluation metrics to understand a machine learning model's performance on unseen data.

- **Why is model evaluation important?** It ensures the model generalizes well to new data, preventing the deployment of overfitted or underperforming models.
- **What is confusion matrix?** A table used to describe the performance of a classification model, outlining True Positives, True Negatives, False Positives, and False Negatives.
- **Define accuracy, precision, recall, and F1-score.**
  - *Accuracy*: Total correct predictions divided by total predictions.
  - *Precision*: True Positives divided by total predicted positives.
  - *Recall*: True Positives divided by total actual positives.
  - *F1-score*: The harmonic mean of precision and recall.
- **What is AUC-ROC curve?** A performance measurement for classification problems. ROC is a probability curve; AUC represents the degree of separability (ability to distinguish classes).
- **What is holdout method?** Splitting a dataset into two mutually exclusive sets: a training set (e.g., 80%) and a testing set (e.g., 20%).
- **Explain random subsampling.** Repeatedly holding out different random subsets of data for training and testing, and averaging the performance results.
- **What is parameter tuning?** The process of adjusting a model's hyperparameters (settings defined before training, like  $K$  in K-Means) to optimize performance.
- **Difference between training error and testing error.** Training error is measured on the data used to train the model. Testing error is measured on unseen data, which is the true test of model quality.
- **Explain clustering using IRIS dataset.** Passing the petal/sepal measurements into a clustering algorithm (like K-Means) without species labels, to see if the algorithm naturally groups them into three distinct flower types.
- **What are false positives and false negatives?** *False Positive (Type I error)*: Model incorrectly predicts the positive class. *False Negative (Type II error)*: Model incorrectly predicts the negative class.
- **What is sklearn.metrics?** A module in Scikit-learn that implements functions to assess prediction errors, such as `accuracy_score` and `mean_squared_error`.
- **How do you interpret clustering results?** By examining the centroids and distribution of data points within each cluster to assign a business or real-world meaning to the grouping.

## Unit VI – Data Visualization and Hadoop

- **What is Data Visualization?** The graphical or pictorial representation of data and information.
- **Why is visualization important in Data Science?** The human brain processes visual information much faster than spreadsheets. Visuals easily highlight trends, outliers, and patterns.
- **What are the types of data visualization?** Charts (bar, pie), graphs (line, scatter), plots (box, histogram), and interactive dashboards.
- **Explain line plot.** A graph that displays information as a series of data points connected by straight line segments. Used heavily for time-series data.
- **What is scatter plot?** A graph that uses dots to represent values for two different numeric variables, ideal for observing correlation.
- **Explain histogram.** A visual representation of the frequency distribution of continuous data, grouped into continuous bins.
- **What is density plot?** A smoothed, continuous version of a histogram that estimates the probability density function of a continuous variable.
- **What is box plot?** A standardized way of displaying the distribution of data based on a five-number summary: minimum, first quartile, median, third quartile, and maximum.
- **Difference between histogram and bar graph.** Histograms represent continuous quantitative data (no gaps between bars). Bar graphs represent categorical data (gaps between bars).
- **What tools are used for data visualization?** Tableau, Power BI, Matplotlib, Seaborn, Plotly, and D3.js.
- **What challenges are faced in Big Data visualization?** Overplotting (too many points overlapping), slow rendering speeds, and the difficulty of visualizing high-dimensional data.
- **What is Big Data?** Datasets whose volume, velocity, and variety are too large and complex to be processed by traditional database management tools.
- **Explain Hadoop ecosystem.** A suite of open-source tools and frameworks (like HDFS, MapReduce, YARN, Hive, Pig) designed to store and process big data.
- **What is Hadoop?** An open-source framework developed by Apache that allows for the distributed processing of large datasets across clusters of computers using simple programming models.
- **What is MapReduce?** A programming model for processing big data in parallel. "Map" filters and sorts data; "Reduce" summarizes the mapped data.
- **Explain Mapper and Reducer.** The Mapper processes input data and generates intermediate key-value pairs. The Reducer aggregates those intermediate pairs to produce the final output.

- **What is Apache Pig?** A high-level platform for creating MapReduce programs used with Hadoop. Its scripting language is called Pig Latin.
- **What is Apache Hive?** A data warehouse software project built on top of Hadoop that facilitates reading, writing, and managing large datasets using a SQL-like interface called HiveQL.
- **Difference between Pig and Hive.** Pig is procedural and better for programmers building data pipelines. Hive is declarative (SQL-like) and better for data analysts running queries.
- **What are advantages of Hadoop?** Highly scalable, cost-effective (uses commodity hardware), fast, and fault-tolerant (replicates data).
- **Explain analytical techniques in Big Data visualization.** Aggregating data, using heatmaps instead of thousands of scatter points, and data sampling to reduce visual clutter without losing trends.
- **How is Python used for visualization?** Python utilizes specialized libraries like Matplotlib and Seaborn to convert arrays and DataFrames directly into visual charts via simple code commands.
- **What is matplotlib?** The foundational 2D plotting library for Python, allowing users to create publication-quality figures.
- **What are outliers in box plots?** Individual data points drawn as dots that fall well outside the typical distribution (specifically, beyond the "whiskers" or  $1.5 \times \text{IQR}$ ).
- **Explain visualization using IRIS dataset.** Creating scatter plots to see how sepal length correlates with sepal width, or using a pairplot to visualize the separability of the three flower species.
- **What is distributed processing?** Breaking down a massive computation task into smaller chunks that are executed concurrently across multiple networked computers.
- **What are the components of Hadoop ecosystem?** HDFS (Storage), YARN (Resource Management), MapReduce (Processing engine), and auxiliary tools like Hive, Pig, and Spark.
- **Difference between traditional databases and Hadoop.** Traditional DBs use "schema-on-write", scale vertically, and handle structured data. Hadoop uses "schema-on-read", scales horizontally, and handles all data types.
- **What is scalability in Hadoop?** It features horizontal scalability, meaning you can increase capacity simply by adding more servers (nodes) to the cluster without downtime.
- **What are limitations of Hadoop?** Not suitable for real-time data processing (it is batch-oriented), has a steep learning curve, and managing small files is inefficient.

## **Practical 1 – Data Wrangling I (Short Viva Q&A)**

1. **What is Data Wrangling?**  
Cleaning and transforming raw data into usable format.
2. **What is Pandas?**  
Python library for data analysis.
3. **What is DataFrame?**  
Tabular data structure in Pandas.
4. **Use of isnull()?**  
Checks missing values.
5. **Use of describe()?**  
Shows statistical summary.
6. **What is normalization?**  
Scaling data to a common range.
7. **What is Label Encoding?**  
Converting categories into numbers.
8. **What is One-Hot Encoding?**  
Creating binary columns for categories.
9. **What are categorical variables?**  
Variables containing labels/categories.
10. **What is preprocessing?**  
Preparing raw data for analysis.

## **Practical 2 – Data Wrangling II (Short Viva Q&A)**

1. **What is an outlier?**  
Abnormal data value.
2. **What is IQR?**  
Difference between Q3 and Q1.
3. **What is skewness?**  
Measure of data asymmetry.
4. **What is log transformation?**  
Technique to reduce skewness.
5. **What is fillna()?**  
Fills missing values.
6. **What is dropna()?**  
Removes missing values.
7. **Why use median?**  
Less affected by outliers.

8. **What is Q-Q Plot?**  
Checks normal distribution.
9. **What is KDE plot?**  
Smooth distribution curve.
10. **What is data transformation?**  
Converting data into suitable form.

### **Practical 3 – Descriptive Statistics (Short Viva Q&A)**

1. **What is mean?**  
Average value.
2. **What is median?**  
Middle value.
3. **What is mode?**  
Most frequent value.
4. **What is variance?**  
Measure of spread.
5. **What is standard deviation?**  
Square root of variance.
6. **What is percentile?**  
Value below certain percentage.
7. **What is dispersion?**  
Spread of data.
8. **What is central tendency?**  
Center value of data.
9. **What is groupby()?**  
Groups data category-wise.
10. **What is Iris dataset?**  
Flower classification dataset.

### **Practical 4 – Linear Regression (Short Viva Q&A)**

1. **What is Linear Regression?**  
Predicts continuous values.
2. **What is dependent variable?**  
Output variable.
3. **What is independent variable?**  
Input variable.

4. **What is RMSE?**  
Prediction error measure.
5. **What is  $R^2$  score?**  
Model accuracy indicator.
6. **What is residual?**  
Actual – predicted value.
7. **What is overfitting?**  
Model memorizes training data.
8. **What is train-test split?**  
Dividing dataset for training/testing.
9. **What is correlation?**  
Relationship between variables.
10. **What is Boston Housing dataset?**  
House price prediction dataset.

## **Practical 5 – Logistic Regression (Short Viva Q&A)**

1. **What is Logistic Regression?**  
Binary classification algorithm.
2. **What is sigmoid function?**  
Converts output into probability.
3. **What is confusion matrix?**  
Table of prediction results.
4. **What is accuracy?**  
Correct predictions percentage.
5. **What is precision?**  
Correct positive predictions ratio.
6. **What is recall?**  
Detected actual positives ratio.
7. **What is TP?**  
True Positive.
8. **What is FP?**  
False Positive.
9. **What is feature scaling?**  
Scaling features to same range.
10. **What is classification?**  
Predicting categories/classes.

## Practical 6 – Naive Bayes (Short Viva Q&A)

1. **What is Naive Bayes?**  
Probability-based classifier.

2. **What is Bayes theorem?**

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

3. **Why called “Naive”?**  
Assumes features are independent.

4. **What is Gaussian Naive Bayes?**  
Assumes normal distribution.

5. **What is prior probability?**  
Probability before prediction.

6. **What is posterior probability?**  
Probability after evidence.

7. **What is likelihood?**  
Probability of feature given class.

8. **What is multiclass classification?**  
More than two classes.

9. **Which dataset used?**  
Iris dataset.

10. **Advantage of Naive Bayes?**  
Fast and simple.

## Practical 7 – Text Analytics (Short Viva Q&A)

1. **What is NLP?**  
Natural Language Processing.

2. **What is tokenization?**  
Splitting text into words.

3. **What are stopwords?**  
Common unnecessary words.

4. **What is stemming?**  
Reducing word to root form.

5. **What is lemmatization?**  
Converting word to meaningful root.

6. **What is POS tagging?**  
Identifying grammar category.



7. **What is TF?**  
Term Frequency.
8. **What is IDF?**  
Inverse Document Frequency.
9. **What is TF-IDF?**  
Importance measure of words.
10. **What is text preprocessing?**  
Cleaning textual data.

## **Practical 8 – Data Visualization I (Short Viva Q&A)**

1. **What is data visualization?**  
Graphical representation of data.
2. **What is histogram?**  
Distribution graph.
3. **What is Seaborn?**  
Python visualization library.
4. **What is Titanic dataset?**  
Passenger survival dataset.
5. **Purpose of histogram?**  
Shows frequency distribution.
6. **What is Matplotlib?**  
Plotting library in Python.
7. **What is countplot?**  
Shows category counts.
8. **What is fare column?**  
Ticket price.
9. **Why visualize data?**  
To identify patterns/trends.
10. **What is EDA?**  
Exploratory Data Analysis.

## **Practical 9 – Data Visualization II (Short Viva Q&A)**

1. **What is box plot?**  
Shows data distribution and outliers.
2. **What is median in boxplot?**  
Middle line inside box.

3. **What are whiskers?**  
Range indicators.
4. **How are outliers shown?**  
Points outside whiskers.
5. **What is quartile?**  
Division of data into four parts.
6. **What is IQR?**  
 $Q3 - Q1$ .
7. **Why use boxplots?**  
To detect outliers/spread.
8. **What is categorical data?**  
Label-based data.
9. **What is numerical data?**  
Number-based data.
10. **What inference means?**  
Conclusion from analysis.

## **Practical 10 – Data Visualization III (Short Viva Q&A)**

1. **What is feature distribution?**  
Spread of feature values.
2. **What is histogram used for?**  
Checking distribution.
3. **What is boxplot used for?**  
Detecting outliers.
4. **What are nominal variables?**  
Non-numeric categories.
5. **What are numeric variables?**  
Number-based variables.
6. **What is outlier detection?**  
Finding abnormal values.
7. **Why compare distributions?**  
To understand feature behavior.
8. **What is Iris flower dataset?**  
Flower measurement dataset.
9. **What are feature types?**  
Numeric and categorical.
10. **What is visualization inference?**  
Insights obtained from graphs.

All the best

- Karan salunkhe 😊